

Improving Methodological Research in Psychology: Preregistration, Transparent Reporting, and Benchmarking

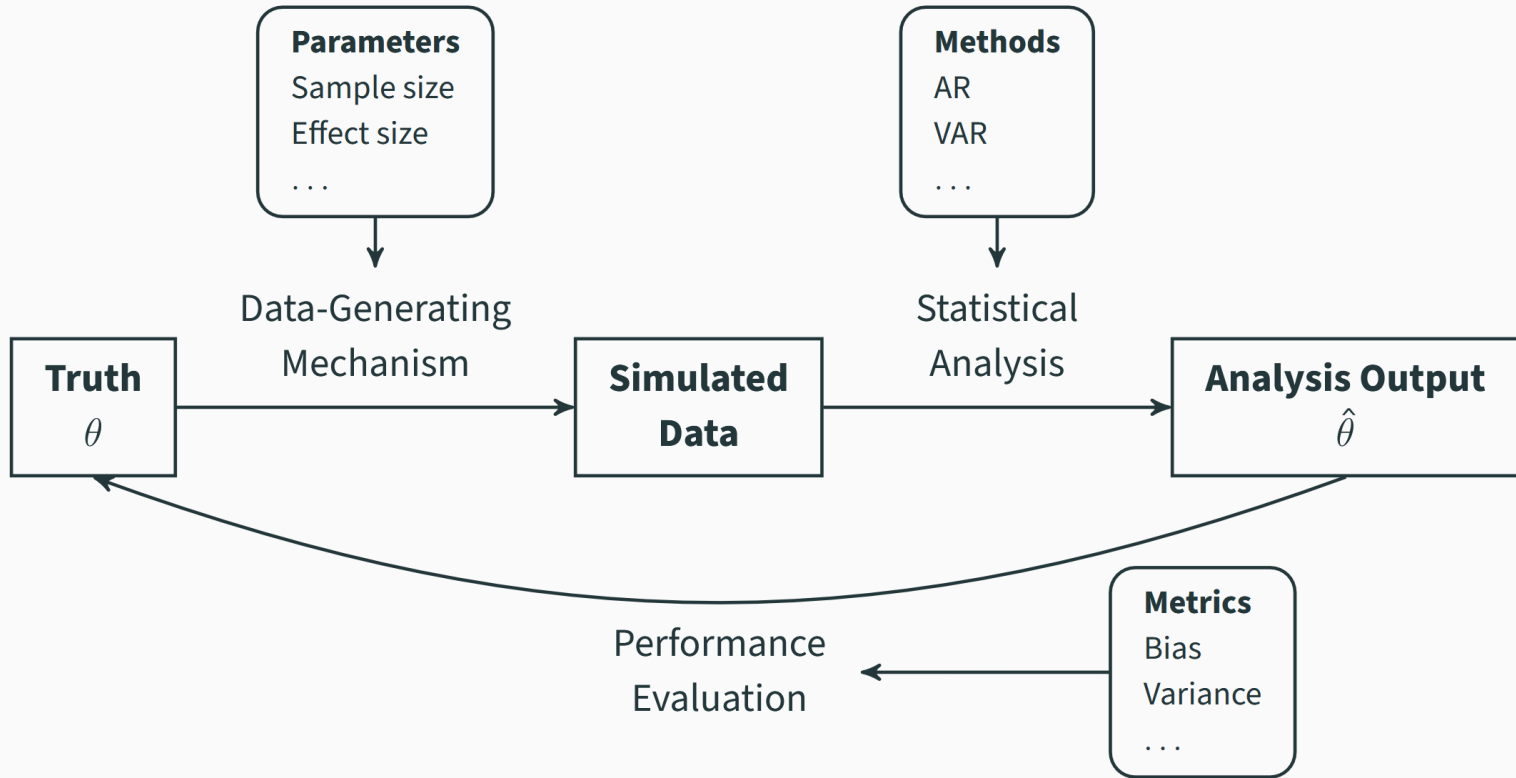
Leibniz-Institut für Bildungsverläufe · June 2026

Björn Siepe

Psychological Methods Lab, Department of Psychology, Marburg University

Part I: Background and Issues

Simulation studies



Issues in simulation studies



Marcin Wichary - Wikimedia

“ *Statisticians ... often pay too little attention to their own principles of design*
Hoaglin & Andrews, 1975

Meta-Science on Simulation Studies

Interdisciplinary

- (Bio-)statistics: Morris et al. (2019)
- Ecology: Williams et al. (2024)
- Psychometrics: Feinberg & Rubright (2016)

Transparency

- Issues similar to other empirical research (Boulesteix et al., 2020)
- Insufficient reporting standards (e.g., Hoaglin and Andrews, 1975)
- Reproducibility? (e.g., Luijken et al., 2023)

Execution Issues

- Many degrees of freedom (Pawel et al., 2024)
- Over-Optimism (e.g., Ullmann et al., 2022)
- Unclear handling of non-convergence (Pawel et al., 2025; Wünsch et al., 2025)



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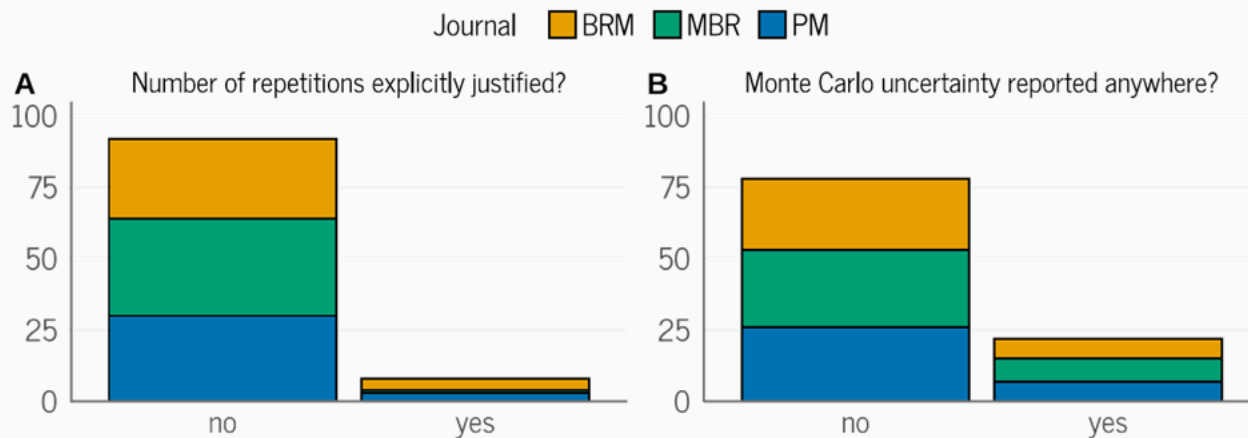
Psychological Methods

<https://doi.org/10.1037/met0000695>

Simulation Studies for Methodological Research in Psychology: A Standardized Template for Planning, Preregistration, and Reporting

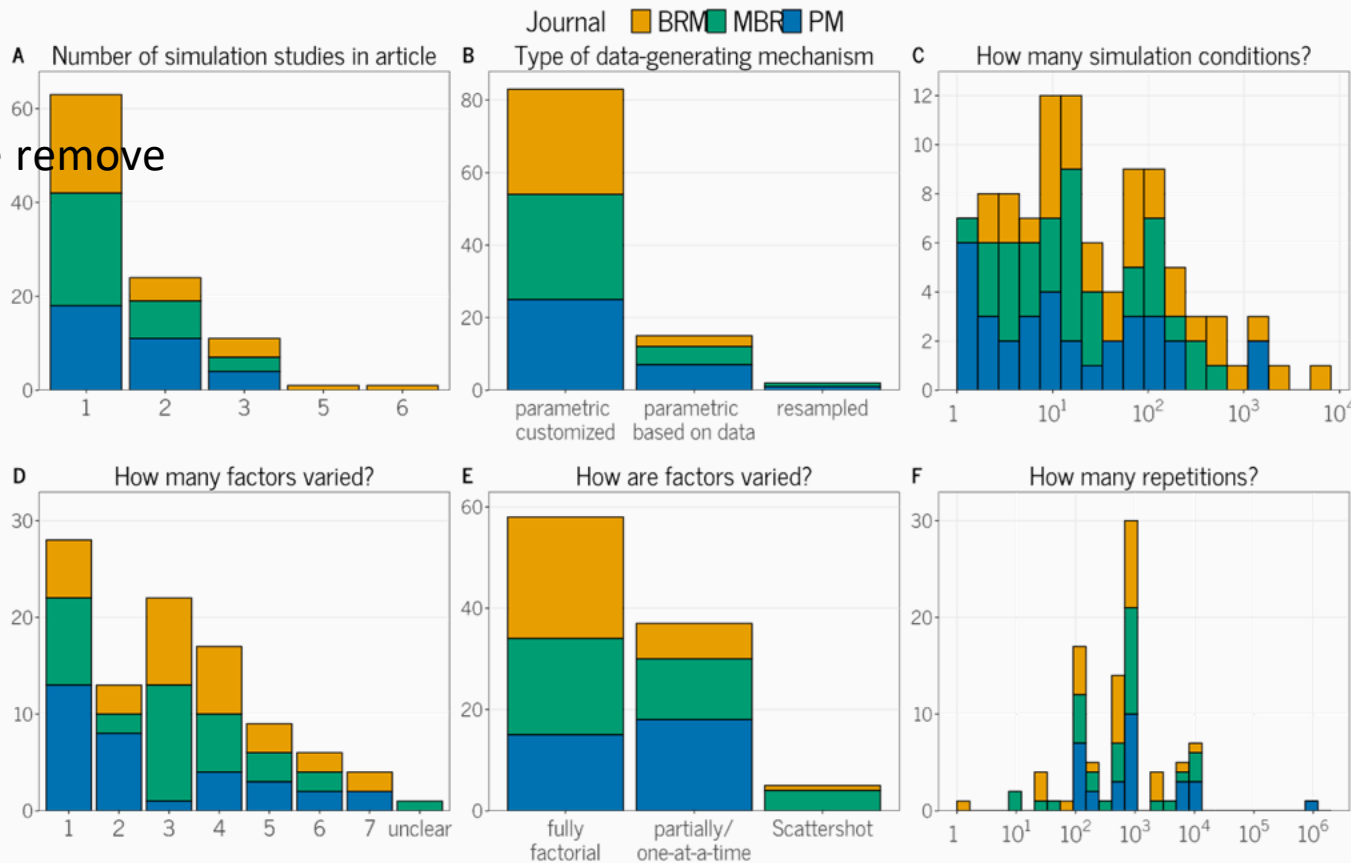
Björn S. Siepe¹, František Bartoš², Tim P. Morris³, Anne-Laure Boulesteix^{4, 5},
Daniel W. Heck¹, and Samuel Pawel^{6, 7}

Simulation studies in psychology

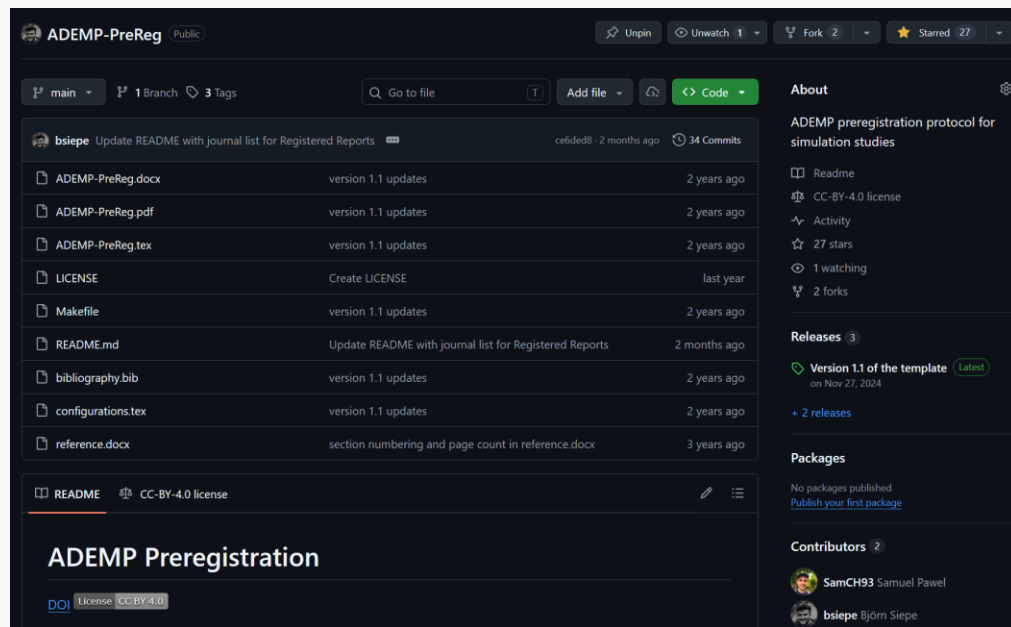


Simulation studies in psychology

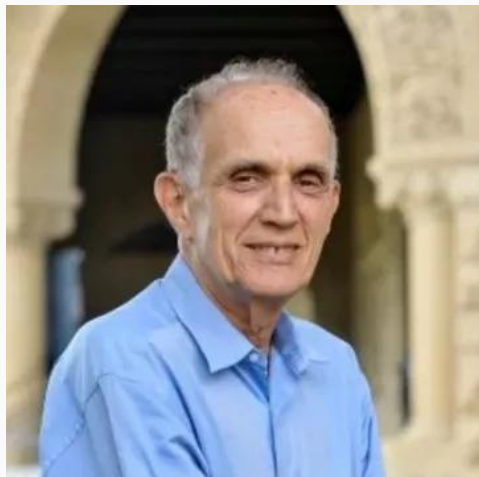
Maybe remove



1. Instructions
2. General information
3. **Aims**
4. **Data-generating mechanisms**
5. **Estimands & targets**
6. **Methods**
7. **Performance measures**
8. **Computational details**



Issues in simulation studies



<https://statistics.stanford.edu/people/bradley-efron>

“[...] it is *very difficult to run an honest simulation comparison, and easy to inadvertently cheat by choosing favorable examples, or by not putting as much effort into optimizing the dull old standard as the exciting new challenger.*

Brad Efron, 2001

Potential Solutions

Adversarial Collaboration



The Splintered Mind Blog

Analysis Blinding



Artwork by Sandbox Studio, Chicago with Corinne Mucha

Reporting Standards/Full Disclosure/Lab Handbook

**ADEMP-PreReg
Template for Simulation Studies**

Part II: To preregister or not to preregister?

The idea of simulation protocols



Research Article

The design of simulation studies in medical statistics

[Andrea Burton](#) ✉, [Douglas G. Altman](#), [Patrick Royston](#), [Roger L. Holder](#)

First published: 31 August 2006 | <https://doi.org/10.1002/sim.2673> | [VIEW METRICS](#)

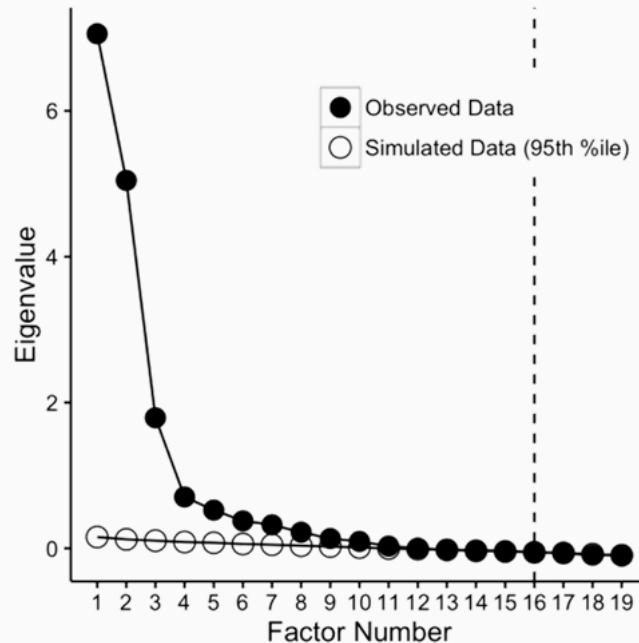
“When planning a simulation study, it is recommended that a detailed protocol be produced, giving full details of how the study will be performed, analysed and reported.”

Burton et al., 2006

Claims based on simulation studies

Add
picture
on EFA

Add
citation
to our
paper



Sakaluk & Short (2016)

“*We recommend parallel analysis for ordinal items*

Hypothetical authors

Illustration of preregistration

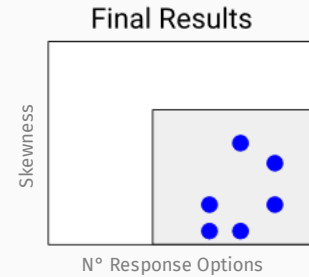
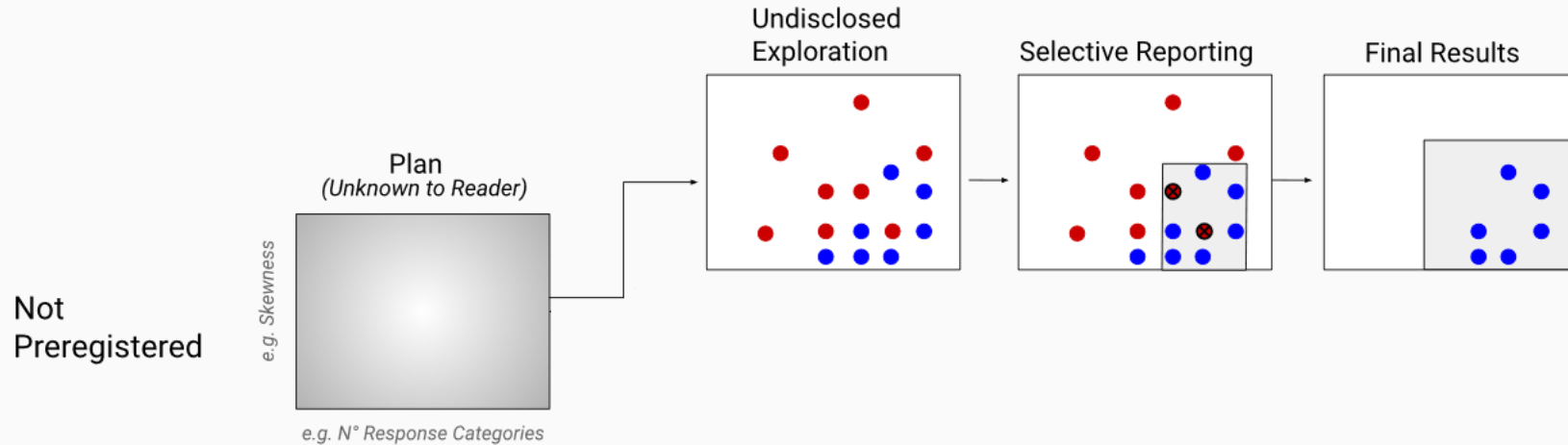


Illustration of preregistration



Theoretical advantages of preregistration



Preregistration reduces **epistemic uncertainty** about the process that produced simulation results

Falsificationist

Simulation only counts as evidence if it could have caught the method failing

Inductionist

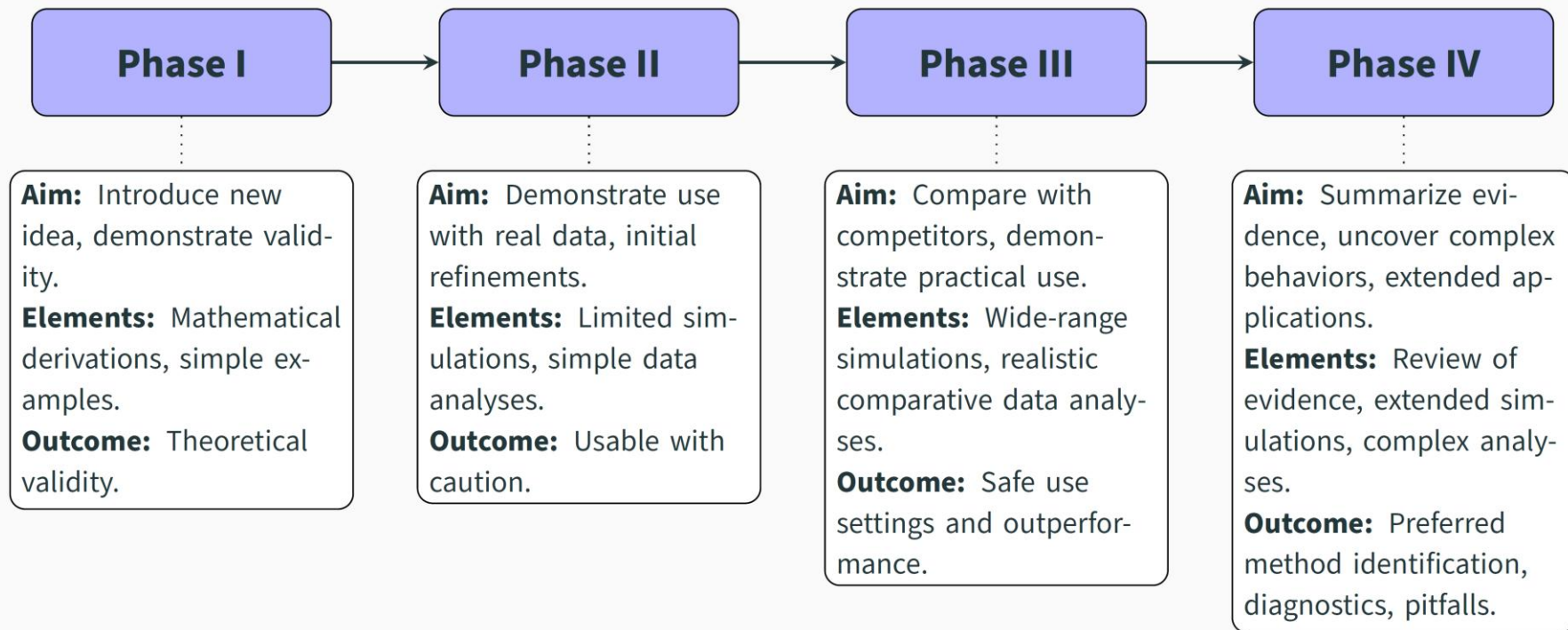
Convincing simulations need difficult conditions (catch a bad method) and easy enough conditions (let a good method succeed)

Predictivist

Predicting an outcome and prespecifying a plan in advance carries more evidential weight than explaining it afterward

When might they work?

Increasing usefulness of preregistration →



FAQ about preregistration



Is preregistration in itself a sign of good quality?

What about deviations?

Do I need theoretical expectations?

When does a simulation study start?

What about cheating?

No, but allows assessment thereof

Be transparent

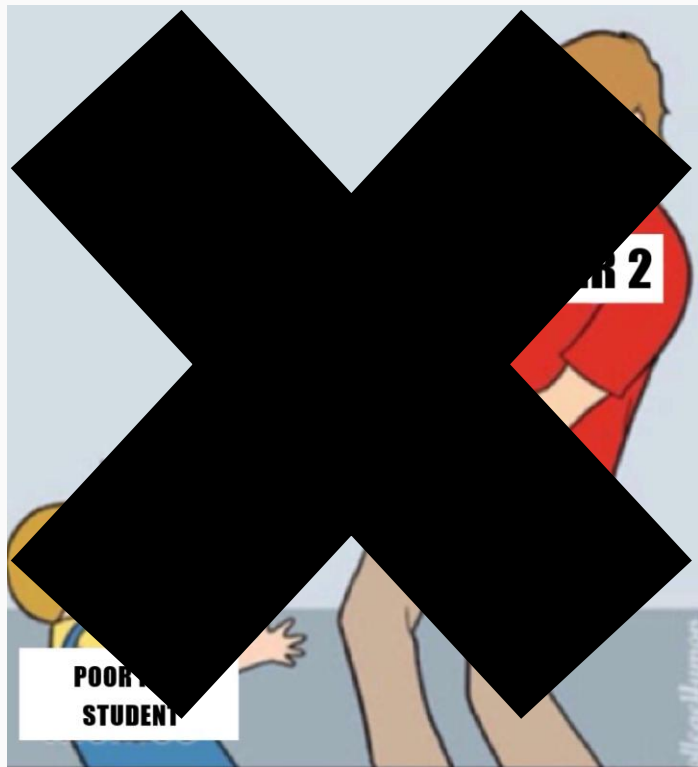
No, but can be helpful

Hard to determine, but there are practical tools

Can never be fully ruled out



Registered Reports to the rescue



Created with ImageResizer

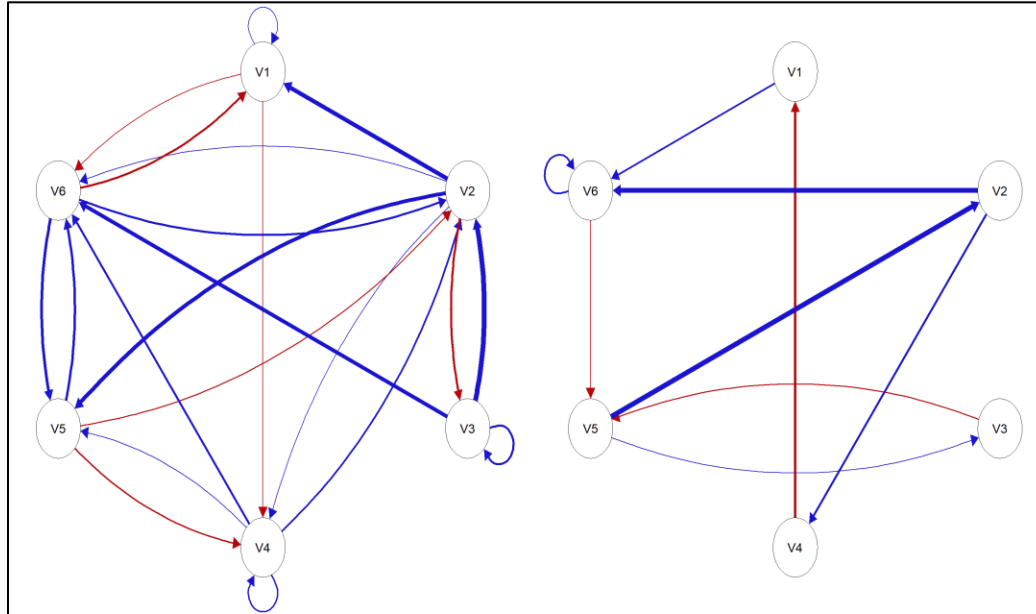


Centre for Open Science

Part III: Benchmarking

Particular Issues in Psychology

What is a realistic data-generating mechanism?



Hard to understand when performance differences actually matter

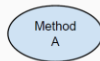
Living Synthetic Benchmarks: A Neutral and Cumulative Framework for Simulation Studies

František Bartoš ¹, Samuel Pawel ², Björn S. Siepe ³

Incentivizing cumulative simulation research

Separate Studies (Status Quo)

Paper 1
(new method)



Continuous Synthetic Benchmarking (Proposal)

DGM: Data-Generating Mechanism
PM: Performance Measure

Synthetic benchmarking for simulation studies

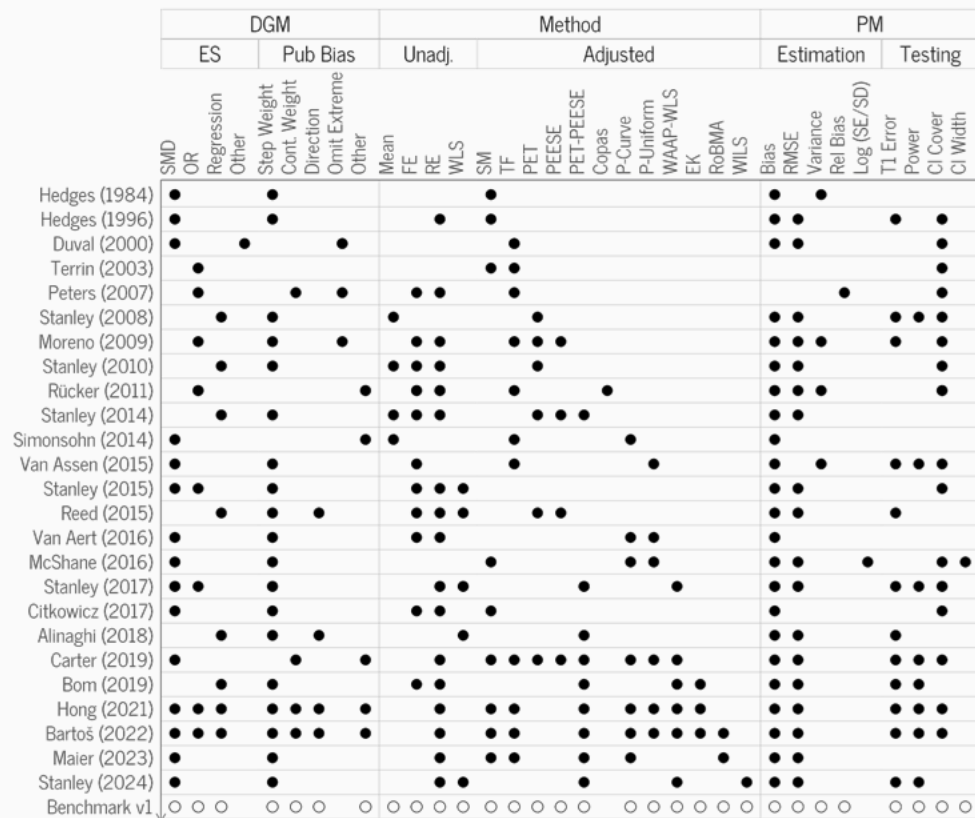


Figure 3: Timeline of major simulation studies on publication bias adjustment

Synthetic benchmarking for simulation studies

fbartos.github.io/PublicationBiasBenchmark/

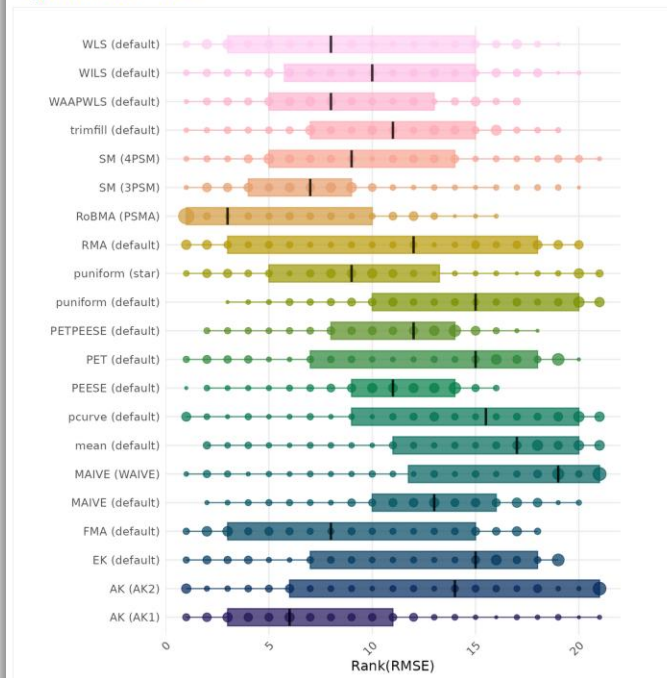
By-Condition Performance (Conditional on Method Convergence)

The results below are conditional on method convergence. Note that the methods might differ in convergence rate and are therefore not compared on the same data sets.

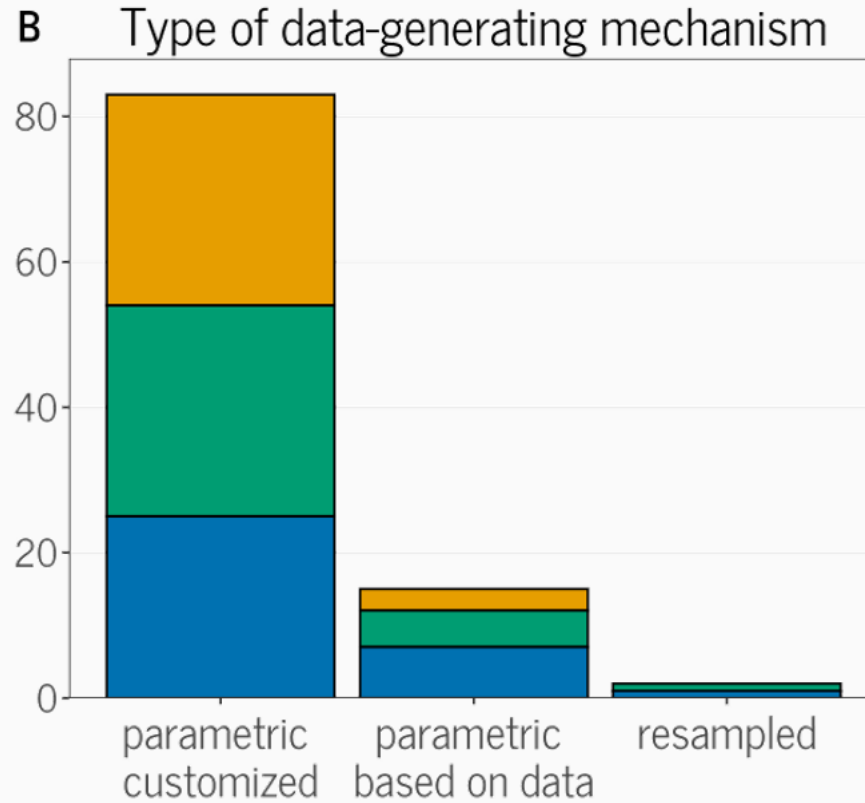
Convergence **RMSE** Bias Empirical SE Interval Score 95% CI Coverage

95% CI Width Log Positive Likelihood Ratio Log Negative Likelihood Ratio

Type I Error Rate Power



What about *real* benchmarking?



Real-data benchmarking

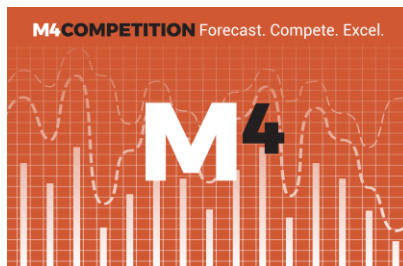
Idea: Application of different algorithms to real-world data

Computer Vision/ML



Source: Steni Sebastian, Medium

Forecasting



Source: Rob Hyndman

Biostatistics

Robinson and Vitek *Genome Biology* (2019) 20:205
<https://doi.org/10.1186/s13059-019-1846-5>

EDITORIAL

Benchmarking comes of age

Mark D. Robinson^{1*} and Olga Vitek^{2*}

Psychology has been slow to adopt benchmarking (Rocca & Yarkoni, 2021)

Introducing openESM



Database

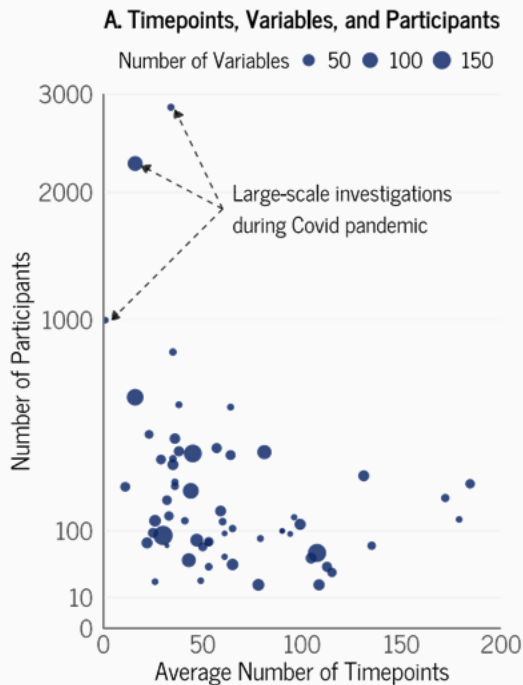
Open-source
Harmonized
Rich metadata

Scale

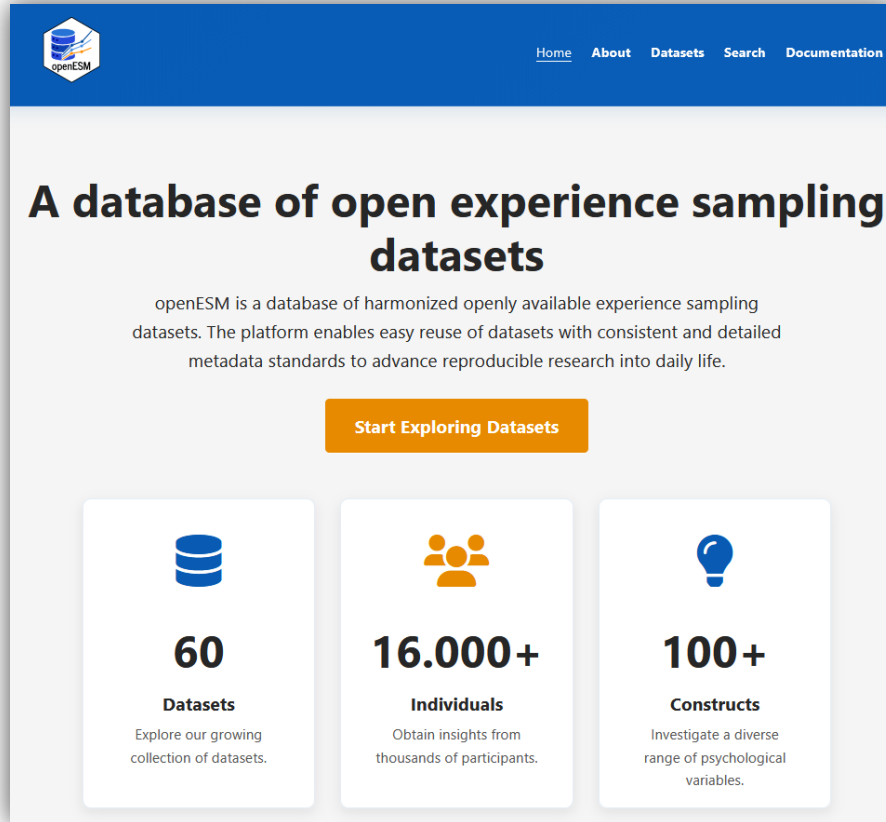
60 datasets
>16,000 participants
>740,000 observations

Access

Website with search
 and  packages



Siepe, B. S., Haslbeck, J. M. B., Kloft, M., Büchner, A., Zhang, Y., Fried, E. I., & Heck, D. W. (2025)



The screenshot shows the homepage of the openESM website. At the top is a blue navigation bar with the openESM logo on the left and links for Home, About, Datasets, Search, and Documentation on the right. The main content area has a light gray background. It features a large heading 'A database of open experience sampling datasets' followed by a paragraph describing the database. Below this is an orange button labeled 'Start Exploring Datasets'. At the bottom, there are three white cards, each with an icon, a large number, a title, and a short description.

openESM

[Home](#) [About](#) [Datasets](#) [Search](#) [Documentation](#)

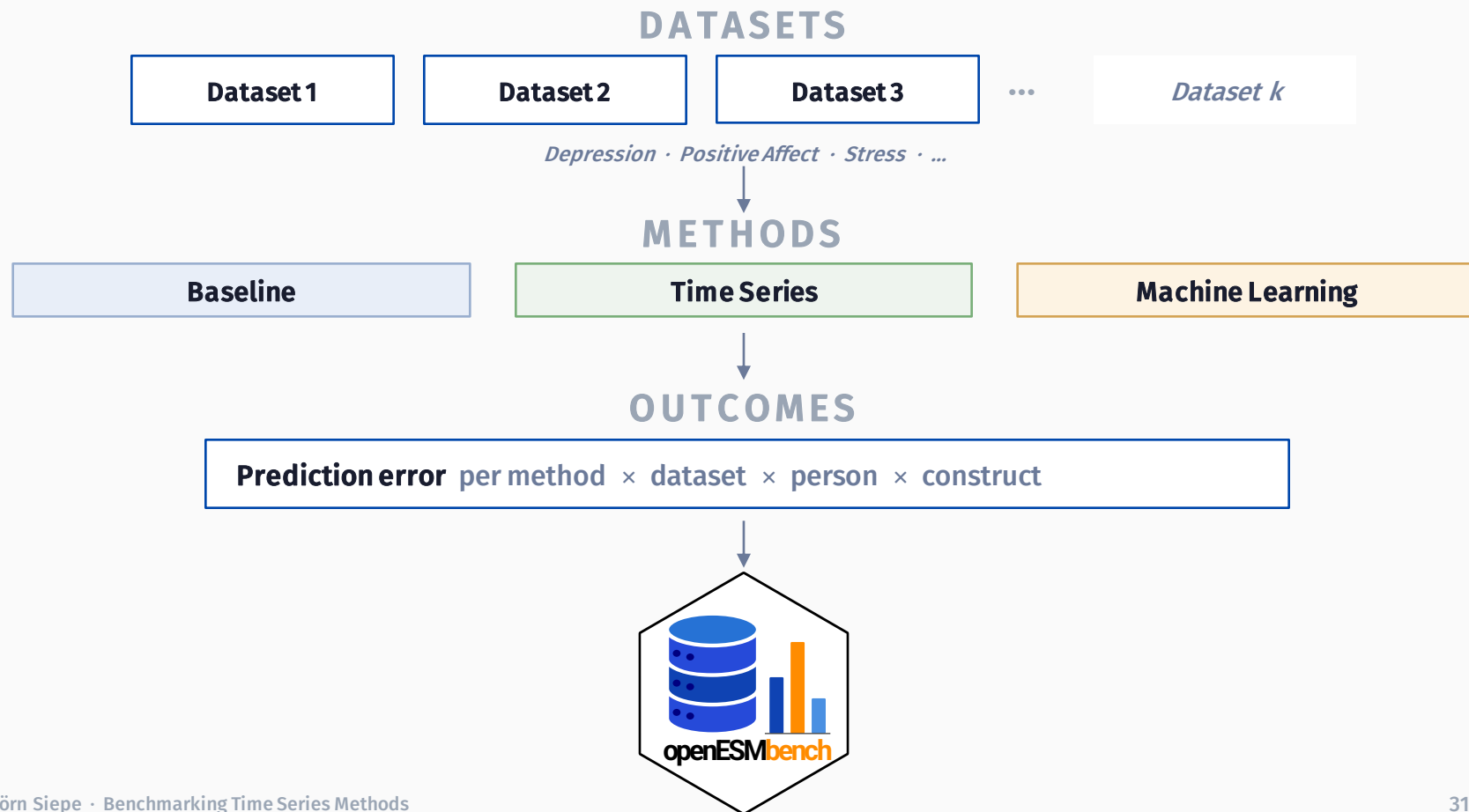
A database of open experience sampling datasets

openESM is a database of harmonized openly available experience sampling datasets. The platform enables easy reuse of datasets with consistent and detailed metadata standards to advance reproducible research into daily life.

[Start Exploring Datasets](#)

Icon	Count	Category	Description
	60	Datasets	Explore our growing collection of datasets.
	16.000+	Individuals	Obtain insights from thousands of participants.
	100+	Constructs	Investigate a diverse range of psychological variables.

Benchmarking psychological time series methods



Methodological research has **quality issues** like any empirical field

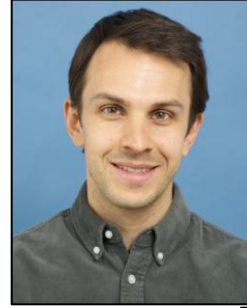
Two potential solutions:

Preregistration
makes generalizability
assessable and process
transparent, **not**
automatically better

Benchmarking
closes the gap in **two ways**:

- explicit conditions
- real-world grounding

Collaborators



Thank You!



<https://openesmdata.org>



bjoernsiepe@gmail.com



<https://bsiepe.github.io>



bsiepe.bsky.social

Slides & Papers



References

Efron, B. (2001). Statistical modeling: The two cultures: Comment. *Statistical Science*, 16(3), 218-219.

Robinson, M.D., Vitek, O. (2019). Benchmarking comes of age. *Genome Bio*/20, 205 <https://doi.org/10.1186/s13059-019-1846-5>

Rocca, R., & Yarkoni, T. (2021). Putting Psychology to the Test: Rethinking Model Evaluation Through Benchmarking and Prediction. *Advances in Methods and Practices in Psychological Science*, 4(3), 10.1177/25152459211026864.
<https://doi.org/10.1177/25152459211026864>

Sakaluk, J. K., & Short, S. D. (2016). A Methodological Review of Exploratory Factor Analysis in Sexuality Research: Used Practices, Best Practices, and Data Analysis Resources. *Journal of Sex Research*.

Siepe, B. S., Haslbeck, J. M. B., Kloft, M., Büchner, A., Zhang, Y., Fried, E. I., & Heck, D. W. (2025). Introducing openESM: A database of openly available experience sampling datasets. https://doi.org/10.31234/osf.io/qfdtb_v1

Resources

- All icons either from PowerPoint or fontawesome
 - <https://fontawesome.com/icons/bluesky>
 - <https://fontawesome.com/icons/glasses>
 - <https://fontawesome.com/icons/code>
 - <https://fontawesome.com/icons/dna>
- Python Software Foundation. (n.d.). *The Python logo*. <https://www.python.org/community/logos/> (PSF Trademark Policy)
- Rob Hyndman - <https://robjhyndman.com/hyndsight/m4comp/>
- Steni Sebastian - <https://medium.com/@ssteni/imagenet-what-why-and-how-e5ebed04abb5>
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- Carbon (<https://carbon.now.sh/>) for pretty code screenshots
- The Splintered Mind - <https://schwitzsplinters.blogspot.com/2021/02/adversarial-collaboration.html>
- Sandbox Studio Artwork - https://www.symmetrymagazine.org/article/the-facts-and-nothing-but-the-facts?language_content_entity=und
- Registered Reports graphic: Centre for Open Science, https://cdn.cos.io/media/images/registered_reports.width-800.png

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